From Retinal Flow to Predictive Visual Fields: Image-Native Language in Continuous State and Pixel Space

Imagized Language Model Project

August 2026

Abstract

Most language models represent language through symbolic tokens and treat visible writing as an auxiliary modality. We study a stricter object: an independent model whose inputs, state supervision, probability model, and primary outputs are images of writing. We implement the Retinal Flow Language Model (RFLM), a causal model that reads ordered 32×32 image fixations through a continuous retina, integrates them in a recurrent visual field, scores arbitrary candidate images with an energy function, writes the next fixation with conditional pixel-space rectified flow, and rereads its own generated ink. The 11.69M parameter implementation contains no tokenizer, Unicode IDs, OCR transcript, character output table, discrete visual codebook, or external language-model call. V6 adds training on image trajectories sampled by the deployed writer. V7 adds normalized full-history advantage against independent image anchors and differentiable identity learning through sampled pixel-flow endpoints. On a frozen 512-character, four-view Chinese benchmark, selected V7 full-context top-1 rises from V6's 1.20% to 2.31%, exceeding last-only (2.02%) and unigram (1.86%); generated context cosine gain rises from +0.0077 to +0.0303. Nevertheless, normalized target-log-probability gain remains negative (−0.2155), top-1 remains far below the symbolic bigram (13.58%), and the matched 32-cell output is unreadable. We reject V7 as a language model. The failure motivates a Predictive Visual Field: first model a distribution over next image-derived retinal states, then use a separate pixel actuator to render a sampled state and reread it. This is a falsifiable next hypothesis, not a present capability.

1 Motivation

Human readers do not see byte-pair tokens. They see pages, lines, strokes, spacing, damage, font variation, and script-specific visual forms. Current language models are extremely capable, but their central representation is normally a sequence of text tokens. Multimodal models accept images, and modern APIs also expose image-generation endpoints. For example, OpenAI documents APIs for processing images as input and generating images as output, and Google's Gemini API documentation describes image generation from text and image prompts [4, 5, 6]. These systems are important baselines, but they do not by themselves define a language model whose primary language substrate is the visible page.

The Imagized Language Model (ILM) goal is different:

$$\text{image input} \rightarrow \text{visual language state} \rightarrow \text{image output.}$$

The output should itself be a readable answer image. For a query such as "explain the evolution of character yan", the system should produce a page containing modern explanatory writing, classical-style Chinese, and oracle, bronze, seal, and modern forms. Typed text can still be accepted by a

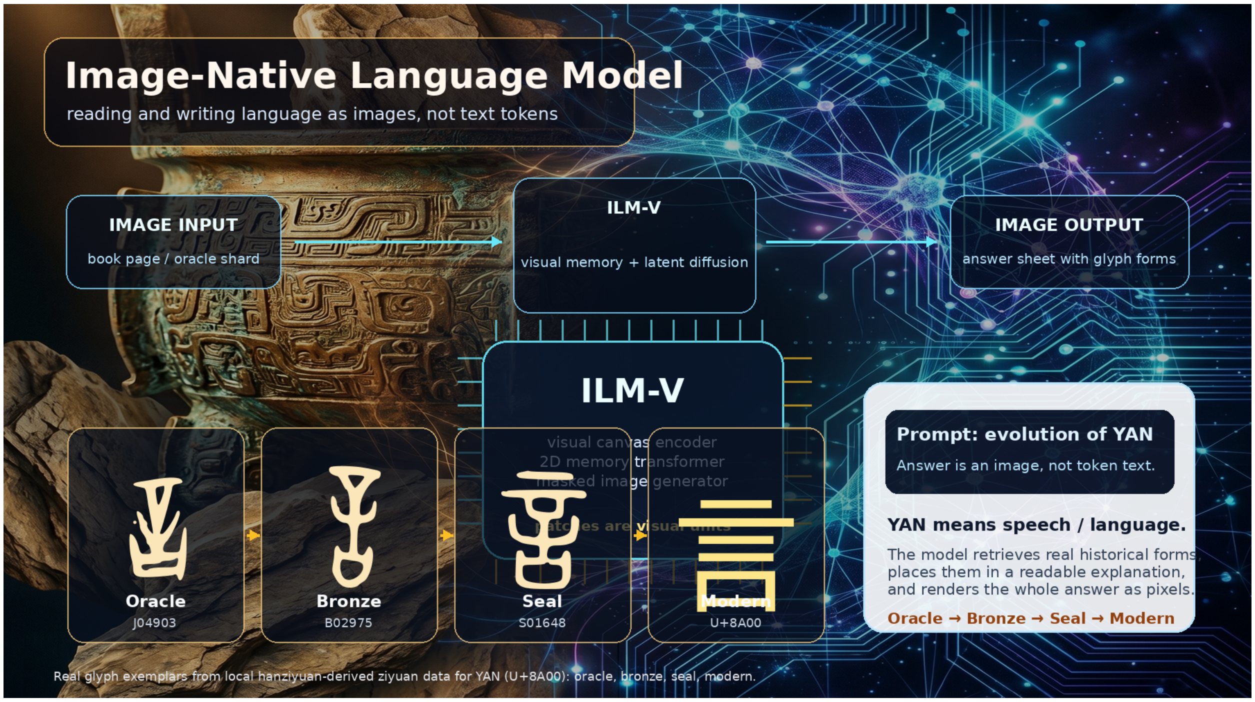

Figure 1: Image-native language modeling concept figure generated with an AgInTi background and overlaid with real local ziyuan glyph exemplars for `YAN` (U+8A00). The answer is a rendered image containing both modern explanation and historical forms.

user interface, but it is rasterized into an image prompt before entering the core model. Conversely, the model's answer is a page image first; OCR or transcription is an optional downstream extraction step.

2 Prior Art and Gap

OCR-free document models such as Donut and screenshot parsing models such as Pix2Struct show that document images can be understood without an explicit OCR pipeline. PIXEL reconstructs masked rendered-text patches and demonstrates cross-script visual transfer [10]; PIXAR is a genuinely generative pixel-space autoregressive language model and identifies noisy maximum-likelihood text images as a central difficulty [11]; PixelGPT studies autoregressive visual and textual pretraining [12]. These are the closest model-level precedents. Token-free language models such as CANINE, ByT5, and MEGABYTE remove subwords but retain symbolic bytes or characters. Discrete diffusion language models show that generation need not be left-to-right, but usually retain a linguistic vocabulary.

I-JEPA shows that predicting target image representations from visual context can learn semantic visual fields without reconstructing every pixel [24]. Flow matching trains continuous generative paths by vector-field regression and supports efficient numerical sampling [23]. Human reading is also sequential and foveated: Chinese readers make saccades and can use both character- and word-related targeting cues [31]. These observations motivate RFLM, but they do not imply biological equivalence.

The remaining gap is not simply "pixels instead of tokens." It is a system whose predictive random variables are continuous writing images, whose memory is learned from ordered visual

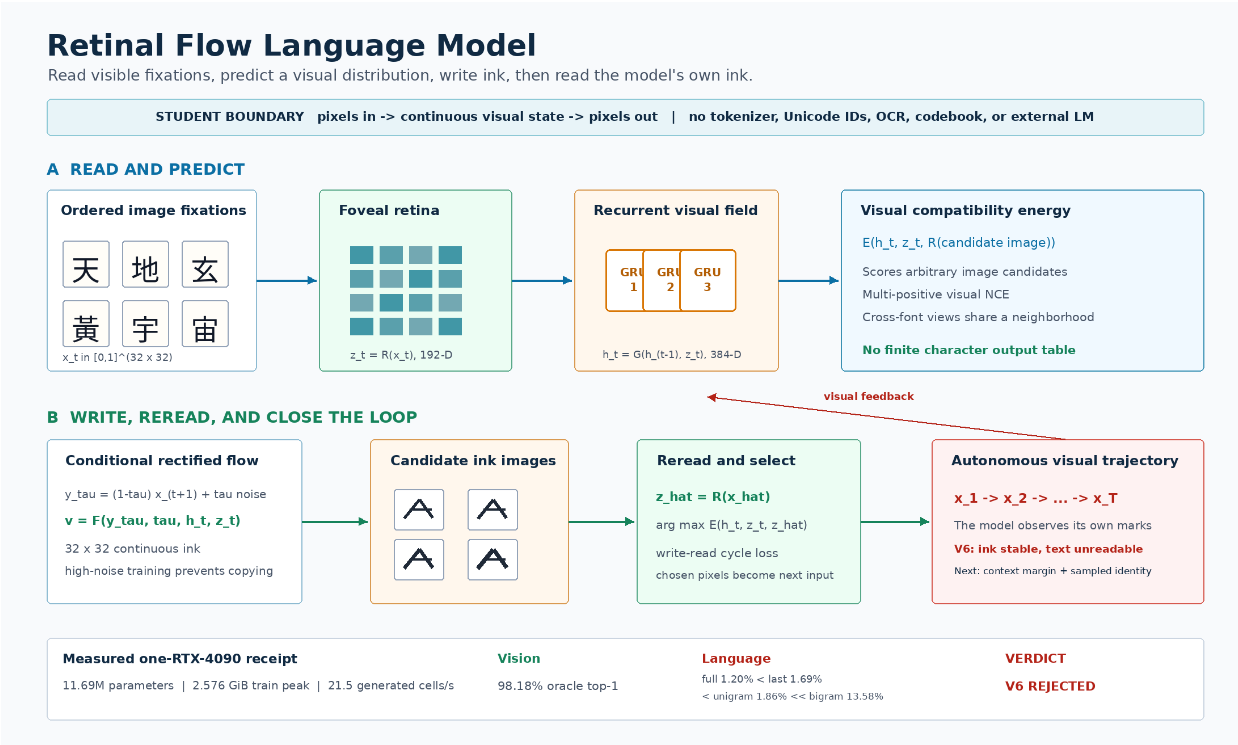

Figure 2: The implemented Retinal Flow Language Model. Training joins visual next-state energy, conditional rectified flow, a write–read cycle, V6 model-induced visual rollouts, and V7 calibrated context and sampled-image identity. Autonomous inference samples candidate ink images, rereads them, selects by visual energy, and feeds the selected pixels back.

observations, whose writer produces new pixels, and whose own output distribution is present during evaluation. Historical evidence adds a separate constraint: synthesized marks must never be presented as attested source glyphs.

3 Architecture

Figure 2 gives the strict visual contract. For fixation images $x_t \in [0,1]^{H\times W}$, RFLM factors a visual sequence as

$$p_\Theta(x_{1:T}) = p(x_1)\prod_{t=1}^{T-1} p_\Theta(x_{t+1} \mid x_{\le t}).$$

The random variables are continuous image fields rather than entries in a linguistic or visual vocabulary.

Foveal retina. A convolutional reader maps each 32×32 image to a normalized 192-dimensional visual state:

$$z_t = R_\theta(x_t)/\|R_\theta(x_t)\|_2.$$

Independent font renderings are aligned with visual contrastive, invariance, and variance objectives. The target retina $R_{\bar\theta}$ is an exponential moving average of the online retina. Font coverage is checked from font cmap tables so missing-glyph boxes cannot become a spurious class.

Recurrent visual field and energy. A three-layer GRU integrates the fixation history,

$$h_t = G_\phi(h_{t-1}, z_t), \qquad h_t \in \mathbb{R}^{384}.$$

An energy function scores any candidate image c_j after rereading it:

$$s_{t,j} = E_\psi(h_t, z_t, R_{\bar{\theta}}(c_j)).$$

Multi-positive visual NCE admits multiple near-identical image views as valid targets. Unlike a softmax language head, this energy is not tied to a finite character inventory.

Conditional ink flow. For clean target x_{t+1}, noise ϵ, and $\tau \in [0,1]$, define

$$y_\tau = (1-\tau)x_{t+1} + \tau\epsilon, \qquad u_\tau = \epsilon - x_{t+1}.$$

The pixel writer predicts the path velocity

$$\hat{u}_\omega = F_\omega(y_\tau, \tau, h_t, z_t), \qquad \mathcal{L}_{\text{flow}} = \|\hat{u}_\omega - u_\tau\|_2^2.$$

Training emphasizes high-noise times, where target pixels cannot simply be copied, and weights sparse strokes and the predicted clean endpoint.

Write–read cycle. The one-step endpoint $\hat{x}_0 = y_\tau - \tau\hat{u}_\omega$ is reread through the frozen target retina. A visual NCE cycle objective aligns it to independent target renderings:

$$\mathcal{L}_{\text{cycle}} = \text{NCE}\Big(R_{\bar{\theta}}(\hat{x}_0), R_{\bar{\theta}}(x_{t+1}^{(1)}), R_{\bar{\theta}}(x_{t+1}^{(2)})\Big).$$

The complete loss combines flow, energy, cross-render invariance, retina contrast, endpoint, stroke, and cycle terms. At inference, the writer samples a candidate set $\mathcal{C}_t$ and selects

$$\hat{x}_{t+1} = \arg\max_{c \in \mathcal{C}_t} E_\psi(h_t, z_t, R_{\bar{\theta}}(c)).$$

The selected pixels become the next observation. This closes the visual loop and turns autonomous stability into a mandatory model property rather than a post-processing concern.

Model-induced visual rollout. V6 invokes the same sampler and energy reranker during training. From a clean prefix it generates two short-flow candidates, selects and detaches one bitmap, rereads it with the online retina, and advances the recurrent state. For two rollout steps it minimizes

$$\mathcal{L}_{\text{roll}} = 0.15\mathcal{L}_{\text{state}} + 0.35\mathcal{L}_{\text{energy}}^{\text{roll}} + 0.30\mathcal{L}_{\text{recovery}}.$$

The terms align clean and induced recurrent states, predict the real next image from the induced state, and recover clean next pixels after generated feedback. All targets remain continuous image fields. Candidate selection is stop-gradient, which bounds memory while exposing the trainable state and writer to deployed pixels.

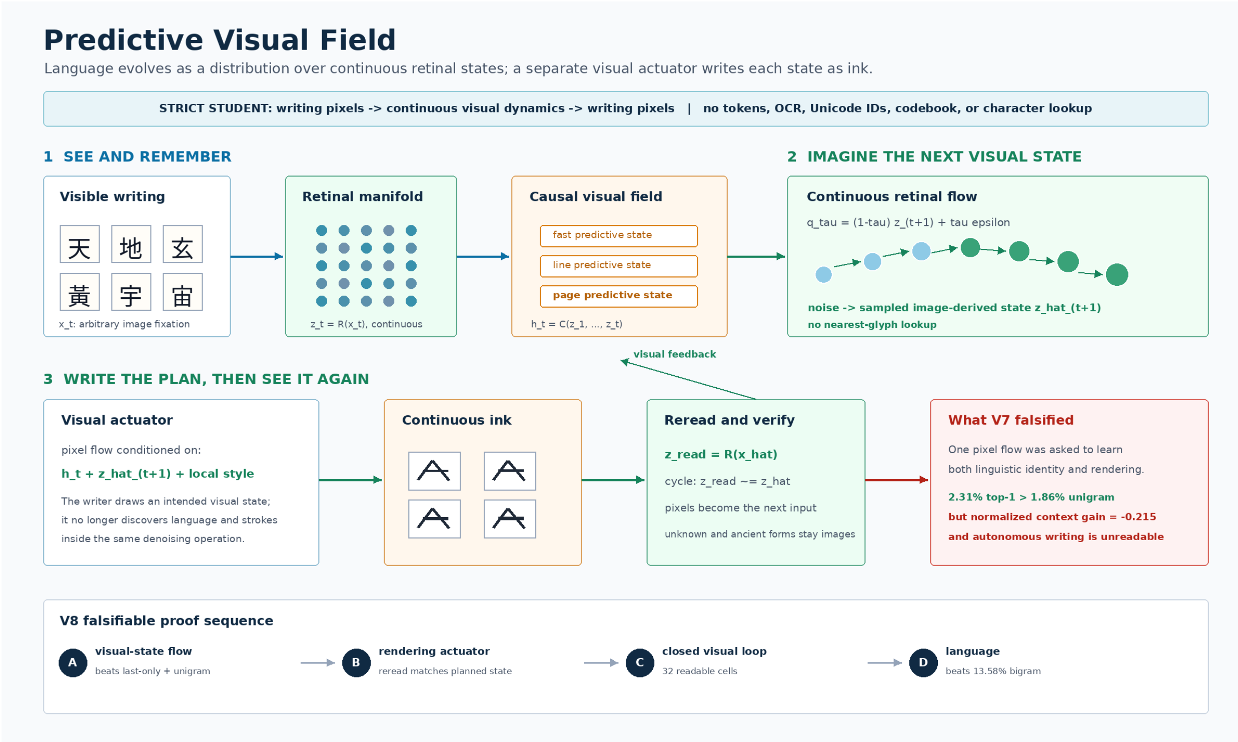

Figure 3: The post-V7 Predictive Visual Field hypothesis. A causal field first models a distribution over continuous image-derived retinal states. A separate visual actuator renders a sampled state, and the generated pixels are reread. V8 must validate state flow and isolated actuation separately before closing the autonomous loop.

Calibrated context and sampled identity. V7 corrects two training/evaluation mismatches. Raw target-energy differences can be increased by changing the scale or all competing scores, so V7 uses the normalized visual target log probability

$$\ell(y \mid h, \mathcal{C}) = s(y \mid h) - \log \sum_{c \in \mathcal{C}} \exp s(c \mid h)$$

and minimizes

$$\mathcal{L}_{\text{ctx}} = \max\{0, m - \ell(y \mid h_{\text{full}}, \mathcal{C}) + \ell(y \mid h_{\text{last}}, \mathcal{C})\}.$$

The last-only branch is detached. The training-only set $\mathcal{C}$ contains independent image renderings; positives are inferred by retinal similarity. Offline strings select the curriculum but anchor IDs and target indices do not enter the student, the pixels are disjoint from evaluation views, and the anchors are not deployed. V7 also differentiates through a two-step numerical pixel-flow endpoint,

$$\mathcal{L}_{\text{sample}} = \text{NCE}\left(R_{\bar{\theta}}(\hat{x}_0^{K=2}), \{R_{\bar{\theta}}(y^{(v)})\}_{v=1}^{V}\right).$$

This trains the sampled image path rather than only a one-step endpoint proxy.

4 Predictive Visual Field Hypothesis

Figure 3 separates linguistic uncertainty from stroke rendering. For target state $z_{t+1} = R_{\bar{\theta}}(x_{t+1})$, Gaussian ϵ, and $\tau \in [0, 1]$, define

$$q_\tau = (1 - \tau)z_{t+1} + \tau\epsilon, \qquad v^* = \epsilon - z_{t+1}.$$

A conditional state field learns

$$\mathcal{L}_{\text{state-flow}} = \|P_\eta(q_\tau, \tau, h_t) - v^*\|_2^2 .$$

Sampling backward from noise yields an intended next visual state $\hat{z}_{t+1}$, not a glyph ID. A separate visual actuator writes

$$\hat{x}_{t+1} = W_\omega(\epsilon_x, h_t, \hat{z}_{t+1}, R_\theta(x_t))$$

and is trained with pixel flow plus a reread constraint

$$\mathcal{L}_{\text{act}} = \mathcal{L}_{\text{pixel-flow}} + \lambda \left[1 - \cos\{R_{\bar{\theta}}(\hat{x}_{t+1}), z_{t+1}\}\right] .$$

No nearest-glyph projection, character classifier, or token unembedding is permitted. This differs from Embedded Language Flows, which demonstrates effective continuous flow dynamics but encodes and finally unembeds discrete tokens [27]. It combines the next-fixation representation objective of recurrent JEPA [25] with the prediction/generation separation explored by D-JEPA [26], while retaining image input and image output.

5 Data

The project already has a strong historical Chinese glyph source outside the tracked repo. A local snapshot at `/home/lachlan/ProjectsLFS/incoder/data/historic` contains:

Item	Count
Characters	9,055
Glyph records	84,642
Oracle	32,216
Bronze	23,500
Liushutong	21,923
Seal	6,996

This supports future evidence-bearing historical answer pages but was not used as a character-label channel in the RFLM run. The language trajectory corpus is a 7,017-record manifest built from 16 public-domain Chinese Wikisource book packages. The contribution layer is recorded as CC BY-SA, and the manifest stores source artifacts and rights. Strings exist only in the offline manifest and renderer; model batches contain image tensors.

Eight Noto Sans/Serif CJK fonts provide independent visual views. The run stores the exact font paths, byte counts, and SHA-256 hashes in checkpoints and receipts. Cmap intersection filtering leaves 8,282 supported visible characters; 107 unsupported unique forms account for only 0.0074% of visible corpus instances and are skipped instead of rendered as missing-glyph boxes.

6 Training Objectives

Each 48-fixation page supplies eight sampled causal positions, making the language objective denser than one endpoint per page. The complete training loss is

$$\mathcal{L} = \lambda_f \mathcal{L}_{\text{flow}} + \lambda_e \mathcal{L}_{\text{energy}} + \lambda_r \mathcal{L}_{\text{retina}} + \lambda_v \mathcal{L}_{\text{view}} + \lambda_p \mathcal{L}_{\text{endpoint}} + \lambda_s \mathcal{L}_{\text{stroke}} + \lambda_c \mathcal{L}_{\text{cycle}} + \rho(t) \mathcal{L}_{\text{roll}}.$$

After step 5,200, V7 adds a ramped correction

$$\mathcal{L}_{\text{V7}} = \mathcal{L} + \kappa(t)\left(0.5\mathcal{L}_{\text{ctx}} + 0.2\mathcal{L}_{\text{sample}}\right).$$

The energy term uses 512 cross-render candidate views per batch in the final run. Visual duplicate positives are discovered by image similarity rather than labels. The view term aligns independent renderings in both retinal and candidate-projection spaces. Endpoint and stroke terms improve sparse-ink geometry. The cycle term is weighted toward high-noise examples, where a writer must use recurrent context rather than visible target pixels.

For V6, $\rho(t)$ ramps from zero to one over 400 updates. A rollout subset of eight sequences uses two generated steps, two candidates, and two flow steps per candidate. This follows the distribution-shift lesson of scheduled sampling and DAgger [29, 28]; it remains image-only because the visited states, selected observations, and recovery targets are pixels and continuous visual fields. The final rollout state cosine reaches 0.961, but selected-image target cosine is only 0.103. This distinction predicts the measured outcome: state recovery becomes robust while sampled visual identity remains weak.

V7 resumes V6 for 800 updates with a 512-form, four-view image-anchor curriculum and a two-step differentiable endpoint on four examples per batch. The anchor retina is refreshed every 200 updates. The selected step 5,800 checkpoint is chosen from preregistered validation checkpoints because it has the strongest combination of independent-anchor context gain (+0.1798) and sampled pixel F1 (0.3636). The continuation adds approximately 261 seconds of training and peaks near 3.0 GiB allocated VRAM on one RTX 4090.

7 Image-First Training and Inference

Figure 4 makes the training contract explicit. Typed corpora are rendered into page images; scanned books and manuscripts are cropped and tiled; historical glyph files become glyph tiles; and marks that are missing from font tables remain valid image patches. Metadata such as crop boxes and stage labels may remain in manifests, but the model path should learn from visual latents. OCR is useful as a readability critic, not as the hidden language channel.

Figure 5 shows the user-facing path. A typed prompt is converted into a visual prompt canvas, and an uploaded image is normalized into the same visual space. The model returns a page-like image containing modern language and historical glyph regions. A later OCR/transcoder can attach a searchable text layer for English, Chinese, or Unicode-representable spans; ancient or unencoded glyphs remain image regions with coordinates and provenance.

The formal contract is:

$$\begin{aligned} r_\alpha(\text{text prompt}) &\to x_{\text{prompt}}, \\ g_\beta(\text{uploaded page or glyph}) &\to x_{\text{prompt}}, \\ f_\theta(x_{\text{prompt}}, x_{\text{context}}) &\to y_{\text{page}}, \\ o_\psi(y_{\text{page}}) &\to \text{optional text layer} + \text{glyph-region metadata}. \end{aligned}$$

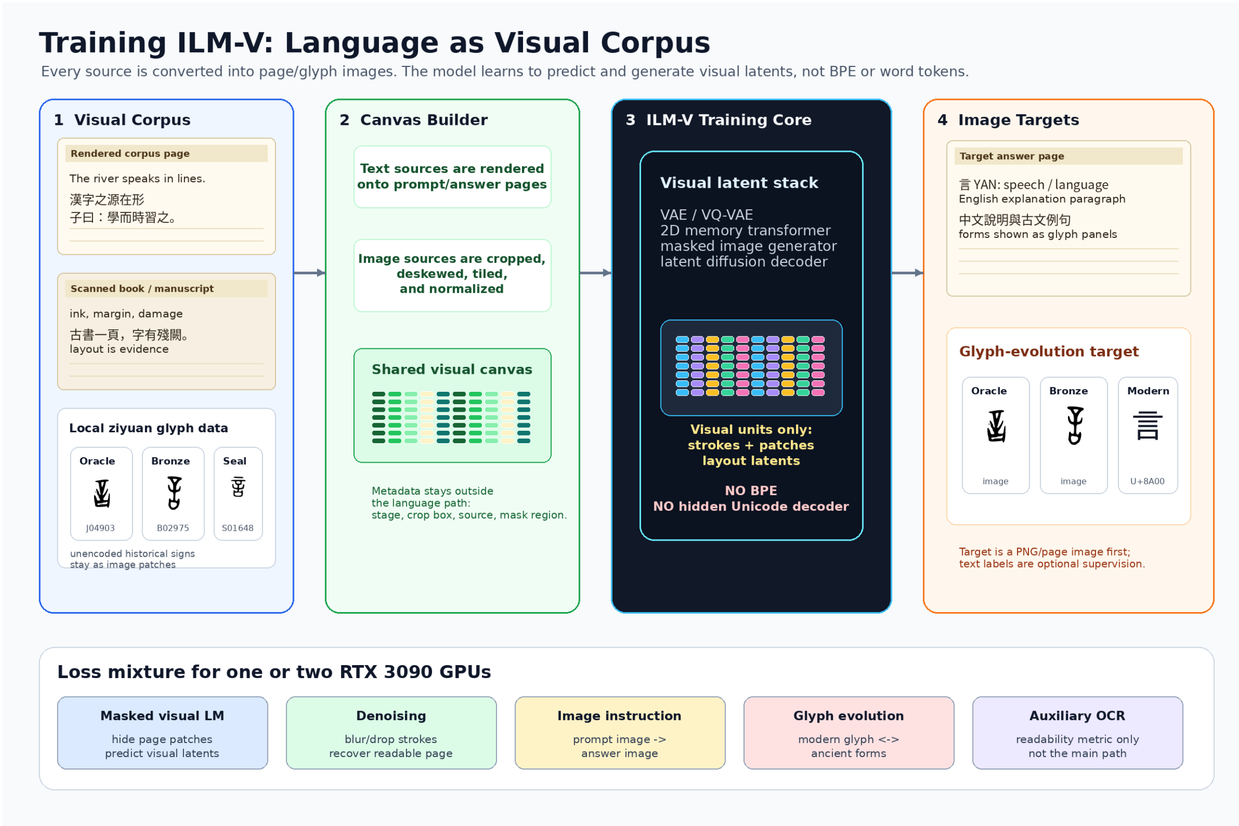

Figure 4: Training pipeline for ILM-V. Rendered text corpora, scanned pages, historical glyph files, and unencoded marks are all converted into visual canvases. The training target is visual latent prediction and image generation, not BPE-token prediction.

Here r_α is only a UI rasterizer, not the language model itself. The core model f_θ maps visual inputs to a rendered page image y_{page}. The optional reader o_ψ may recover Unicode text where possible, but unencoded historical forms remain valid image regions.

A useful answer canvas should resemble a page excerpt from a book or corpus:

- a title region rendered as text image,
- one or more modern-language paragraphs, for example English and Chinese,
- a classical-style line or quotation when appropriate,
- glyph panels for forms that may not exist in computer fonts,
- a provenance strip with source IDs or nearest retrieved exemplars.

This schema is intentionally different from “text answer plus illustrations”. The generated page image is the primary output object, and the text layer is secondary.

8 3090-Friendly Curriculum

Figure 6 records the original page-model schedule. Measurements now support a different order:

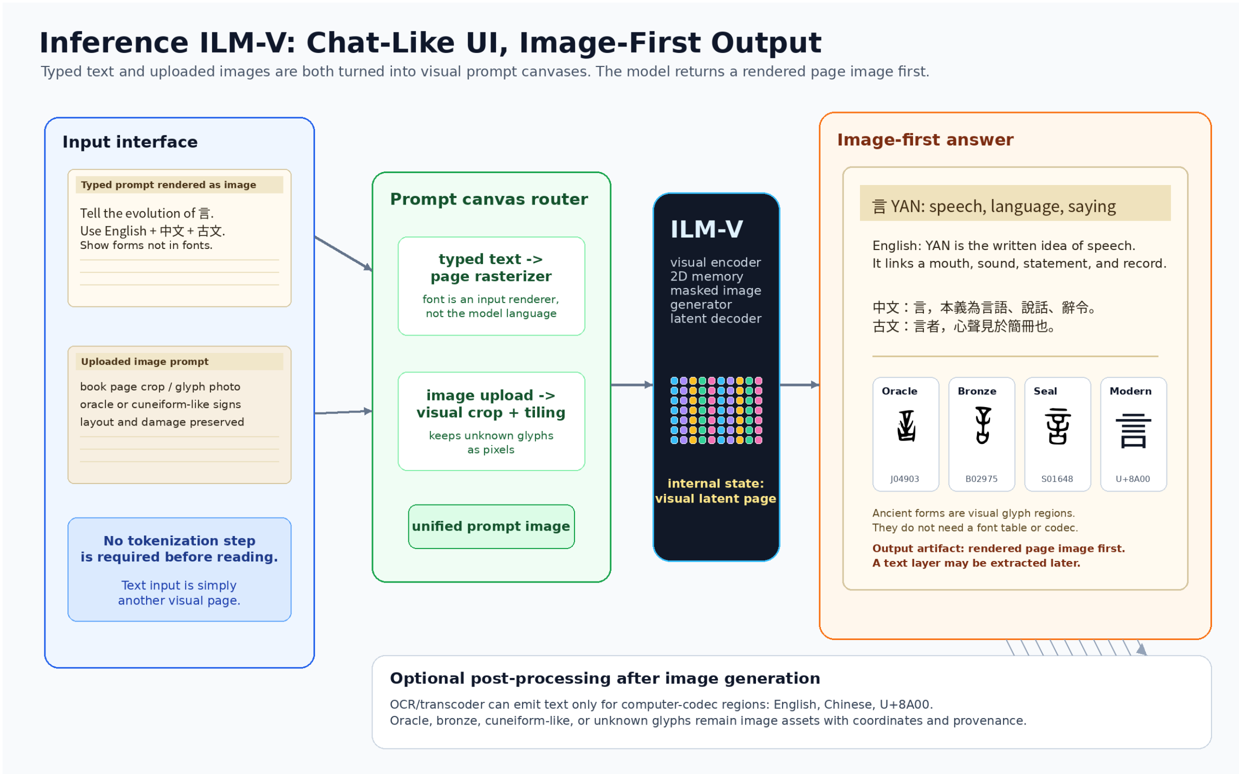

Figure 5: Inference pipeline for ILM-V. A chat-like interface may accept typed text or uploaded images, but typed text is first rasterized into a prompt image. The primary output is a rendered page image. OCR or transcoding is optional and only applies to output regions that exist in computer codecs.

1. Learn a cross-font foveal visual alphabet and audit missing-glyph coverage.

2. Train recurrent visual energy and conditional ink flow jointly on dense causal positions.

3. Reread generated endpoints during training and evaluate on a fixed external visual bank.

4. Train on autonomous model-induced visual trajectories; V6 completes this step and stabilizes ink but remains unreadable.

5. Train normalized context advantage on independent images and identity through differentiable sampled endpoints; V7 improves both signals but remains unreadable.

6. Train continuous next-retina flow alone and require positive normalized context advantage before coupling a renderer.

7. Condition a separate pixel actuator on sampled visual states and validate isolated reread identity before closing the loop.

8. Add line/page timescales, provenance-gated historical panels, and instruction images only after readable autonomous continuation.

9. Scale width and corpus size only after the fixed language and stability gates pass.

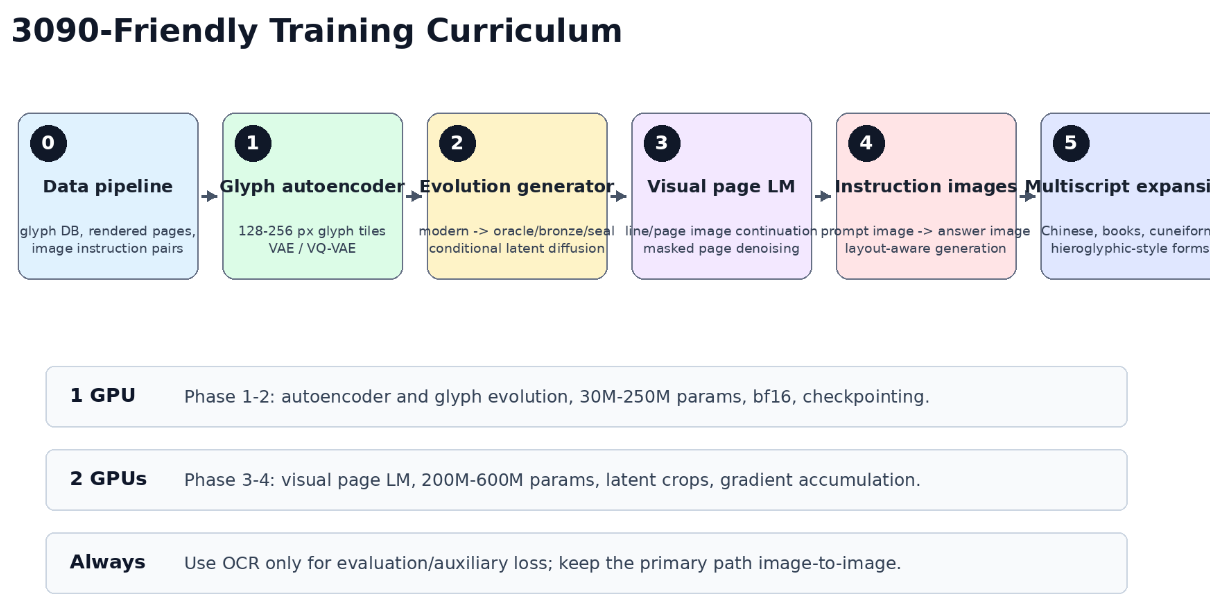

Figure 6: A staged curriculum that fits one or two RTX 3090 GPUs by starting with glyph tiles and latent generation before scaling to page-level image language modeling.

The complete 11.69M model is deliberately small enough for one 24 GiB RTX 4090. Its measured V7 peak near 3.0 GiB leaves room for state flow and a controlled line-state ablation. This is capacity headroom, not an efficiency victory: quality is currently below a symbolic bigram, so “more efficient than an LLM” remains an unsupported hypothesis.

9 Target Demonstration

The first demo should answer a rendered visual prompt such as: “Explain the evolution of the Chinese character yan.” The target answer image should contain:

- a title and concise modern explanation,
- English and Chinese paragraphs rendered as page text,
- a classical-style line such as “yan is the visible trace of speech”,
- a historical timeline with real oracle, bronze, seal, and modern glyph exemplars,
- stage labels and citations as rendered image text.

The existing Zhong figure in Figure 7 is a compact stage-timeline example; the Yan figures in Figures 1 and 5 show the fuller answer-page style. Together they test whether the model can combine language understanding, visual page composition, and historical script generation.

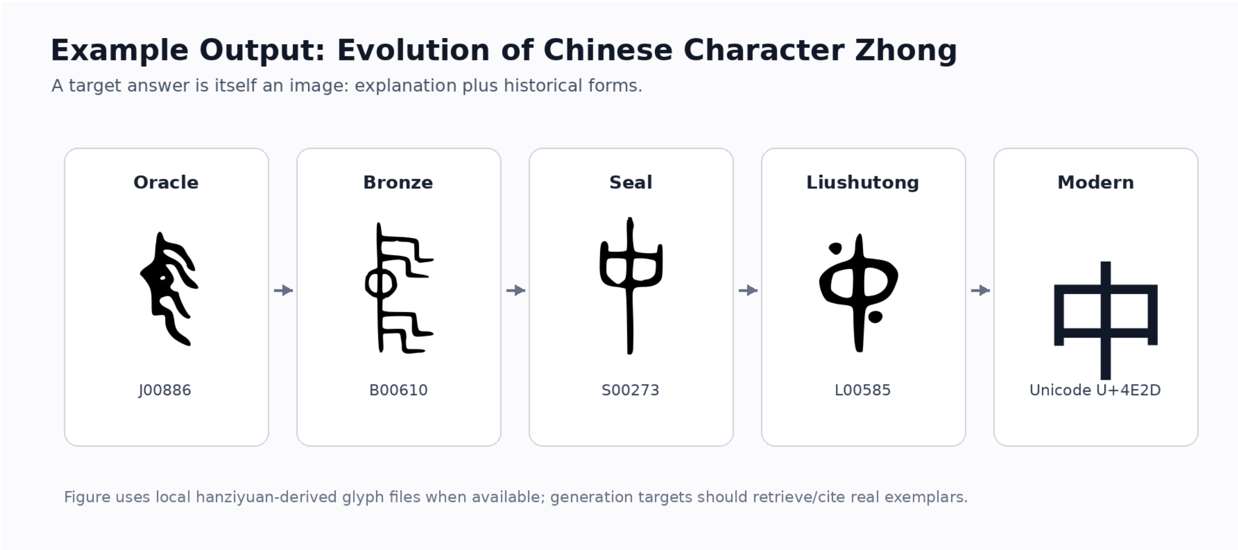

Figure 7: Target output style for a glyph-evolution question. The answer is not merely text: it is a readable image with historical visual forms. The example uses local hanziyuan-derived glyph files when available.

10 AgInTi Artifact Loop

The figure brief for AgInTi-style refinement is stored in the local AgInTi prompts subfolder. The current paper uses scripted PNG generation to keep the artifact reproducible.

11 Evaluation

The model should be evaluated as both a language model and a visual generator:

- Fixed-bank top-1, top-5, and reciprocal rank for full context and a last-fixation ablation.
- Random-weight, chance, unigram, and symbolic bigram baselines on identical contexts.
- Cross-font retina-oracle retrieval, separated from recurrent language prediction.
- Generated-image target similarity, pixel F1, ink occupancy, variance, and energy-reranked hit rate.
- Autonomous rollout length, throughput, VRAM, and blinded human readability.
- Student-boundary audit for token IDs, Unicode IDs, OCR, codebooks, labels, and external models.
- Later historical tasks: glyph-stage accuracy, attested-source provenance, and explicit synthetic-form labeling.

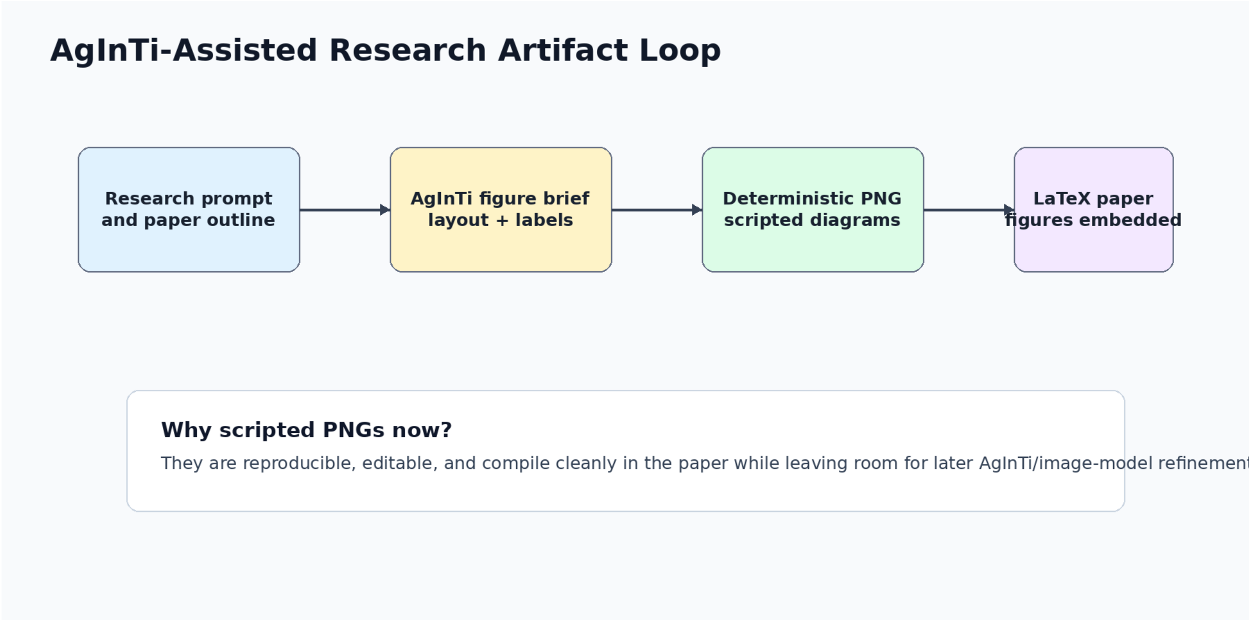

Figure 8: AgInTi-assisted artifact loop. The current figures are deterministic scripted PNGs so the paper is reproducible; the figure brief can later be sent to an interactive AgInTi/image agent for visual refinement.

System	Measured signal	Interpretation
Whole-page flow	velocity MSE 0.851	Unreadable page texture; negative baseline
Global visual memory	top-1 2.23% over 2,016	Above random, but 0/16 historical paraphrases
Causal InkStream	validation ink F1 0.639	Autonomous repeated-grey collapse; negative baseline
RFLM V5 retina oracle	top-1 97.65%	Strong cross-font visual alphabet
RFLM V5 recurrent state	top-1 0.91%	Below last-only 1.36%, unigram 1.86%, and bigram 13.58%
RFLM V6 recurrent state	top-1 1.20%	Improved, but below last-only 1.69% and both language baselines
RFLM V6 pixel writer	context gain +0.0077	Positive but weaker than V5's +0.0211
RFLM V6 autonomous loop	ink ratio 1.168	Stable character-scale marks remain unreadable
RFLM V7 recurrent state	top-1 2.31%	Above last-only 2.02% and unigram 1.86%, far below bigram 13.58%
RFLM V7 calibration	log-probability gain −0.2155	Much better than V6's −0.9066, but full history still lowers mean target probability
RFLM V7 pixel writer	context cosine +0.0303	Passes generated-signal gate; autonomous writing remains unreadable

The fixed evaluation contains 512 common Han candidates with four independent font views

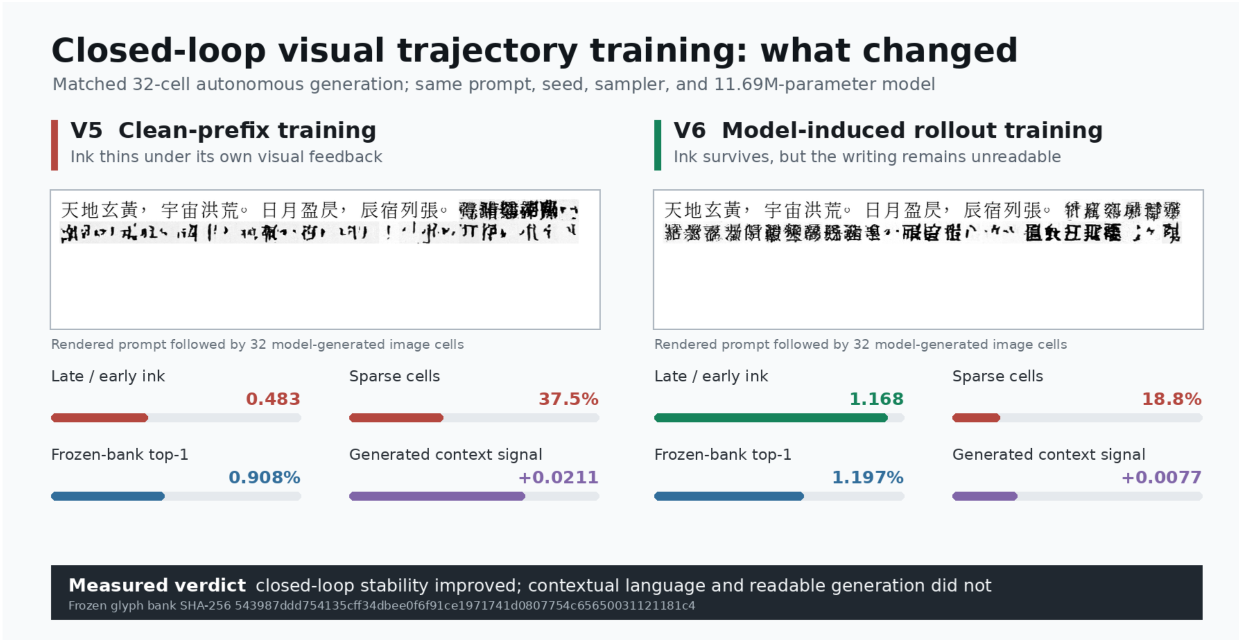

Figure 9: Matched autonomous evidence for V5 and V6. The prompt, seed, sampler, model size, and 32-cell horizon are fixed. Model-induced rollout training prevents the earlier loss of ink and halves the sparse-cell fraction, but neither continuation is readable and generated target signal weakens.

and 2,423 eligible held-out contexts. V6 and V7 use the identical bank SHA-256. Selected V7 raises full-context top-1 from 1.197% to 2.311% and top-5 from 3.219% to 5.613%, while the retina oracle remains near 98.27%. Full context now ranks above V7 last-only (2.022%) and unigram (1.857%) but remains far below symbolic bigram (13.578%). Its normalized target-log- probability gain is −0.2155, improved from a corrected V6 value of −0.9066 but still negative. Raw target-score gain stays positive in both models, showing that raw energy was a misleading metric. The formal acceptance result remains false, and retina-oracle accuracy cannot be cited as language accuracy.

The next proof is staged. First, continuous next-retina flow must beat last-only and unigram and obtain positive normalized context gain without a pixel writer. Second, an isolated actuator must render sampled states with high reread identity and readable ink without lookup. Third, the closed 32-cell loop must remain readable. Beating the symbolic bigram remains the causal-language gate. This order is also consistent with measured error accumulation in iterative diffusion chains [30].

12 Conclusion

The experiments show that merely replacing tokens with pixels is insufficient. A page flow can learn paper statistics, a causal writer can learn local ink, and a retina can learn character identity without acquiring language dynamics. RFLM contributes a strict test harness: ordered continuous visual reading, recurrent image-state prediction, direct conditional ink flow, and rereading under autonomous feedback. V6 further trains on the exact image states produced by its deployed writer. V7 then trains calibrated full-history advantage on independent images and differentiates through sampled endpoints. It crosses unigram and generated-signal gates on one consumer GPU, but

Figure 10: Matched autonomous evidence for V6 and selected V7 step 5,800. V7 improves calibrated retrieval and generated target signal while retaining stable ink, but visual inspection still finds dense unreadable pseudo-glyphs.

negative normalized context probability and unreadable output reject it as a language model. The combined evidence falsifies both exposure-bias correction alone and a monolithic language-plus-rendering pixel writer as sufficient solutions. Predictive Visual Fields make the next test more direct: learn continuous image-derived language dynamics first, then audit a separate renderer and the closed loop in order. Historical etymology answers and broad instruction following remain downstream goals, not present capabilities, and no Qwen-8B parity or efficiency claim is warranted.

References

[1] Brown et al. Language Models are Few-Shot Learners. `https://arxiv.org/abs/2005.14165`

[2] OpenAI. GPT-4 Technical Report. `https://arxiv.org/abs/2303.08774`

[3] OpenAI. GPT-4o API model documentation. `https://developers.openai.com/api/docs/models/gpt-4o`

[4] OpenAI. Images and Vision API documentation. `https://platform.openai.com/docs/guides/images-vision`

[5] OpenAI. Image generation guide. `https://platform.openai.com/docs/guides/image-generation`

[6] Google. Gemini API image generation documentation. `https://ai.google.dev/gemini-api/docs/image-generation`

[7] Meta AI. The Llama 3 Herd of Models. `https://arxiv.org/abs/2407.21783`

[8] Kim et al. OCR-free Document Understanding Transformer. https://arxiv.org/abs/2111.15664

[9] Lee et al. Pix2Struct: Screenshot Parsing as Pretraining for Visual Language Understanding. https://arxiv.org/abs/2210.03347

[10] Rust et al. Language Modelling with Pixels. https://arxiv.org/abs/2207.06991

[11] Tai et al. PIXAR: Auto-Regressive Language Modeling in Pixel Space. https://arxiv.org/abs/2401.03321

[12] Chai et al. Autoregressive Pre-Training on Pixels and Texts. https://arxiv.org/abs/2404.10710

[13] Clark et al. CANINE: Pre-training an Efficient Tokenization-Free Encoder. https://arxiv.org/abs/2103.06874

[14] Xue et al. ByT5: Towards a Token-Free Future. https://arxiv.org/abs/2105.13626

[15] Yu et al. MEGABYTE: Predicting Million-byte Sequences with Multiscale Transformers. https://arxiv.org/abs/2305.07185

[16] Austin et al. Structured Denoising Diffusion Models in Discrete State-Spaces. https://arxiv.org/abs/2107.03006

[17] Li et al. Diffusion-LM Improves Controllable Text Generation. https://arxiv.org/abs/2205.14217

[18] Lou et al. Discrete Diffusion Modeling by Estimating the Ratios of the Data Distribution. https://arxiv.org/abs/2310.16834

[19] Nie et al. Large Language Diffusion Models. https://arxiv.org/abs/2502.09992

[20] Chang et al. MaskGIT: Masked Generative Image Transformer. https://arxiv.org/abs/2202.04200

[21] Rombach et al. High-Resolution Image Synthesis with Latent Diffusion Models. https://arxiv.org/abs/2112.10752

[22] Peebles and Xie. Scalable Diffusion Models with Transformers. https://arxiv.org/abs/2212.09748

[23] Lipman et al. Flow Matching for Generative Modeling. https://arxiv.org/abs/2210.02747

[24] Assran et al. Self-Supervised Learning from Images with a Joint-Embedding Predictive Architecture. https://arxiv.org/abs/2301.08243

[25] Velarde and Parra. Recurrent Joint Embedding Predictive Architecture with Recurrent Forward Propagation Learning. https://arxiv.org/abs/2411.16695

[26] Chen et al. Denoising with a Joint-Embedding Predictive Architecture. https://arxiv.org/abs/2410.03755

[27] Hu et al. ELF: Embedded Language Flows. https://arxiv.org/abs/2605.10938

[28] Ross, Gordon, and Bagnell. A Reduction of Imitation Learning and Structured Prediction to No-Regret Online Learning. `https://proceedings.mlr.press/v15/ross11a.html`

[29] Bengio et al. Scheduled Sampling for Sequence Prediction with Recurrent Neural Networks. `https://papers.nips.cc/paper_files/paper/2015/hash/e995f98d56967d946471af29d7bf99f1-Abstract.html`

[30] Ning et al. Input Perturbation Reduces Exposure Bias in Diffusion Models. `https://proceedings.mlr.press/v202/ning23a.html`

[31] Yan et al. Eye movement guidance in Chinese reading: Is there a preferred viewing location? *Vision Research*, 51(10), 2011. `https://doi.org/10.1016/j.visres.2011.03.004`